BRIDGES IN MATHEMATICS • GRADE 1 • UNIT 3

Adding, Subtracting & Combinations of 10

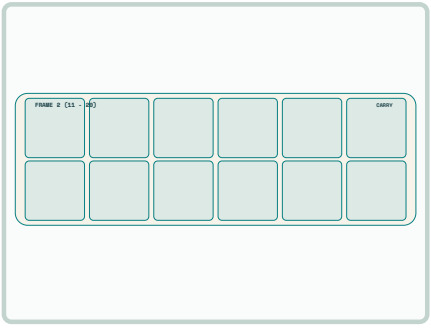
Tactile Concrete-Pictorial-Abstract (CPA) Station Guide • CCSS.MATH.1.OA.B.3

Student Name: _____ Date: _____ **Model: SC-6** **Check Score: ___ / 4**

STUDENT SELF-CHECK RULE (NO CHEATING):
Build with blocks and solve on paper first! Use your SC-6 calculator ONLY to verify answers on the 4-line LCD screen.

STATION LAB TUTORIAL: BENCH SCHEMATIC & CPA PROGRESSION

1. 2D Bench Blueprint



Kit: Ten Frame Shelf

2. Desk Setup (Concrete -> Pictorial)

Physical Bench Setup:
Place 8 discs on shelf 1, 5 discs on shelf 2. Move 2 discs to complete the 10.

Sketch Model on Paper:
Draw two 10-frames with arrows showing making a full ten.

3. Step-by-Step LCD Stack Trace (SC-6)

MODE:	123	Base-10 Arithmetic	
1.	8	X: 8	Key 1st addend (8)
2.	ENTER	Y: 8 | X: 8	Stack lift
3.	2	Y: 8 | X: 2	Key complement (2)
4.	+	X: 10	Complete 10: 8 + 2 =

EXPECTED LCD X: 14 (10 + 4)

1 **Solve 9 + 4 by making 10: 9 + 1 + 3.**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

2 **Solve 8 + 4 by making 10: 8 + 2 + 2.**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

3 **Solve 7 + 6 by making 10: 7 + 3 + 3.**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

4 **Solve 9 + 8 by making 10: 9 + 1 + 7.**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

Station Diagnostic & Solution Key: Unit 3

Adding, Subtracting & Combinations of 10 • Manipulative Setup, Diagnostic Rubric & Verification Flow

MANIPULATIVE HARDWARE STATION INVENTORY & 3D PRINT PREPARATION

Station Kit: Ten-Frame Studio (Dual Frame Shelf)

CAD Blueprint: ten_frame_shelf

Target Device: SC-6 Elementary

CAD / STL Source File:

Education/scad/lib/ten_frame_shelf.scad

Standard: Gridfinity 42mm Pitch / 0.6mm Inset Markings

Recommended Filaments:

Teal Dual Stacking Trays (PLA)

#	Math Problem Prompt	Expected Math Answer	Self-Check Keystrokes & RPN Stack
1	**Solve 9 + 4 by making 10: 9 + 1 + 3.** Standard: CCSS.MATH.1.OA.B.3	13	[9] [ENTER] [1] [+] [3] [+]
2	**Solve 8 + 4 by making 10: 8 + 2 + 2.** Standard: CCSS.MATH.1.OA.B.3	12	[8] [ENTER] [2] [+] [2] [+]
3	**Solve 7 + 6 by making 10: 7 + 3 + 3.** Standard: CCSS.MATH.1.OA.B.3	13	[7] [ENTER] [3] [+] [3] [+]
4	**Solve 9 + 8 by making 10: 9 + 1 + 7.** Standard: CCSS.MATH.1.OA.B.3	17	[9] [ENTER] [1] [+] [7] [+]

TEACHER DIAGNOSTIC GUIDE: COMMON MISCONCEPTIONS & REMEDIATION

Key Misconception to Target During Station Observation:

Watch for students forgetting how many were borrowed from the second addend.

How the Manipulatives & Calculator Foster Self-Correction:

Students solve with pencil and physical blocks first. When they self-check on the SC-6, any discrepancy between their written answer and the calculator's 4-line LCD immediately prompts them to re-inspect their physical bench model or regrouping, turning errors into productive struggle without teacher intervention.

Station Facilitation & Pacing Tip:

Keep 3D printed trays docked on standard Gridfinity baseplates to prevent loose parts on student desks. Encourage students to articulate their CPA steps: name the block representation (Concrete), point to their drawn tape/bond model (Pictorial), and verify register values (Abstract).

CLASSROOM SCORING & STATION MASTERY RUBRIC

Exemplary (18–20 pts):	Complete, accurate manipulative sketches; exact math answers; independently verified self-check on LCD stack.
Proficient (14–17 pts):	Correct math answers; minor sketch omissions; verified on calculator with minimal teacher prompting.
Developing (10–13 pts):	Understands problem concept but made computational slips; successfully self-corrected using calculator LCD.
Intervention (<10 pts):	Relied on calculator before solving on paper, or struggled with underlying physical manipulative models.